CODE BASICS

SQL THROUGH ONE DATASET

DAY 12 / 15

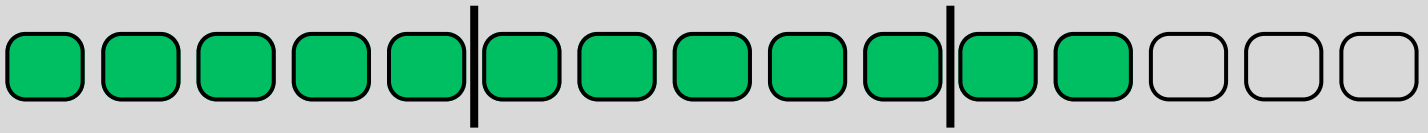
07/08/26

I made a running total reset at fiscal year boundary

One ORDER BY swap fixed whole sequence

ATLIQ HARDWARE GDB0041

MYSQL | Vishnu Ram Sachin | Data Analyst



Problem Statement

DAY 12 / 15

07/08/26

- Yesterday : pivoted fy2020. fy2021 into columns with SUM(CASE())

- Leadership wants a fiscal year build up each months sales everything earlier a running total

The Trap ✕

- ORDER BY fiscal_month_name inside OVER() starts alphabetically , not chronologically

March is evalutated after August

Running total breaks mid fiscal year

Requirment

DAY 12 / 15

07/08/26

reset at every fiscal year ✅

accumulated in true month order (Sep - Aug) ✅

Partition first ,
then order correctly

The Fix

```
SUM(gsr.gross_sales) OVER (
PARTITION BY gsr.fiscal_year
ORDER BY gsr.date ) AS running_total
```

A Quick Analogy

DAY 12 / 15

07/08/26

Car Odometer

Every Trip

All mileage mixed together

(no partition)

Reset Every Year

Odometer rolls backs every september

(PARTITION BY fiscal_year)

Correct order

Trips counted Sep - Aug , not A - Z

(ORDER BY date)

Running Distance

each months odd onto the last

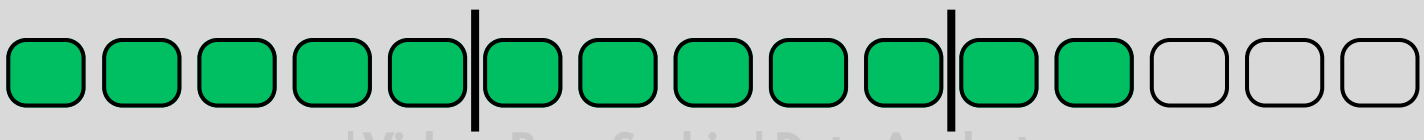
(running_total)

Fix Query

DAY 12 / 15 07/08/26

```sql
WITH gross_sales_report AS (
SELECT fsm.date, fsm.product_code ,
fsm.fiscal_year,
MONTHNAME(fsm.date) AS fiscal_month_name,
fsm.customer_code,
ROUND(SUM(fsm.sold_quantity * fgp.gross_price)/1000000,2) AS gross_sales
FROM fact_sales_monthly fsm
 JOIN fact_gross_price fgp
   ON fgp.product_code = fsm.product_code
   AND fgp.fiscal_year = fsm.fiscal_year
   GROUP BY fsm.fiscal_year , fsm.date
   ORDER BY fsm.fiscal_year,fsm.date )

SELECT
gsr.fiscal_year ,
gsr.fiscal_month_name ,
gsr.gross_sales,
SUM(gsr.gross_sales) OVER (PARTITION BY gsr.fiscal_year
ORDER BY gsr.date ) AS running_total
FROM gross_sales_report gsr
ORDER BY gsr.fiscal_year, gsr.date;
```



Results

DAY 12 / 15

07/08/26

fact_sales_monthly | dim_customer | fact_gross_price | dim_date | SQL File 6*

Limit to 50000 rows

```sql
WITH gross_sales_report AS (
SELECT fsm.date, fsm.product_code , fsm.fiscal_year,
MONTHNAME(fsm.date) AS fiscal_month_name,
fsm.customer_code,
ROUND(SUM(fsm.sold_quantity * fgp.gross_price)/1000000,2) AS gross_sales
FROM fact_sales_monthly fsm
    JOIN fact_gross_price fgp
    ON fgp.product_code = fsm.product_code
```

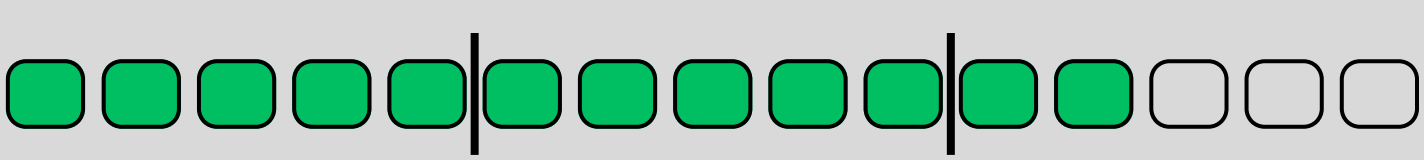
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

fiscal_year	fiscal_month_name	gross_sales	running_total
2018	September	4.19	4.19
2018	October	5.30	9.49
2018	November	7.42	16.91
2018	December	7.54	24.45
2018	January	4.19	28.64
2018	February	4.04	32.68
2018	March	4.46	37.14
2018	April	4.25	41.39
2018	May	4.24	45.63
2018	June	4.13	49.76
2018	July	4.24	54.00
2018	August	4.32	58.32
2019	September	15.14	15.14

SQL BASICS

DAY 12 / 15

07/08/26



| Vishnu Ram Sachin | Data Analyst

SQL THROUGH ONE DATASET CHALLENGE